WORKBOOK 06 / 08 · MULTIMEDIA 12

Give the Caveman a voice

Recording, Stream sound, four mouth poses and true hand swaps

MANUAL FOUR-POSE GRAPHIC BANK



1 · closed | 2 · round | 3 · wide | 4 · teeth

Align visible lip centres. Single Frame holds the selected pose.

The original four mouth drawings, aligned by visible lip centre inside Mouth_Poses.

Build: Synchronize mouth and gesture changes to the actual recording and preserve the original Caveman identity.

Keep: Me_clever Scene + original voice recording + measured cue sheet + lip-sync revision

Shared story: A Caveman wants a third counting stone. A stream stands in his way. He studies it, commits to a jump, retrieves the stone and returns to count his collection with enormous pride.

Document: ActionScript 3.0 · 1280 × 720 · 24 fps · three main Scenes · Graphic symbols for scrubbable character actions

Two provisional 70-minute sessions. Use the HTML guide for read-aloud, switchable supports, recovery files and narrated explanation. This print version includes the essential support prompts.

Continue the same film

Create the project in Workbook 1. For every later workbook, open the selected previous production FLA and Save As a new numbered checkpoint, preserving all three Scenes and the complete Library. Keep repair/director copies separate; return to the chosen production version.

Understand before you animate

Lip sync follows heard sound shapes and phrase emphasis. Gestures support meaning; they do not have to change on every mouth cue.

The original four-mouth method is manual Graphic Single Frame selection: closed, round, wide and teeth. It is a simplified stylized bank, not a full phoneme set. The hand images are three separate Graphic definitions changed with Swap Symbol. The previous 105-frame file held one round mouth; it was a mechanism test, not a completed lip-sync performance. A synthetic test WAV is labelled technical reference only.

Today’s checklist

1. Predict the result. 2. Build four stages. 3. Check in playback. 4. Refine, diagnose and direct. 5. Save and reopen.

Independent director condition

Record a more doubtful or more triumphant reading of the same line. Retain the words, remeasure cues, and redesign the gesture and timing.

What you will defend

Show one audible cue, its timeline frame and the selected mouth pose; then explain one deliberate hold.

Support choices

Use the step summaries, glossary, numbered checklist, different-object examples, explanation frames and optional stretch. Open the companion HTML for browser read-aloud and the narrated six-slide explanation. Supports change access to the task; the success criteria remain the same.

Your planning notes

01 · Record the actual performance first

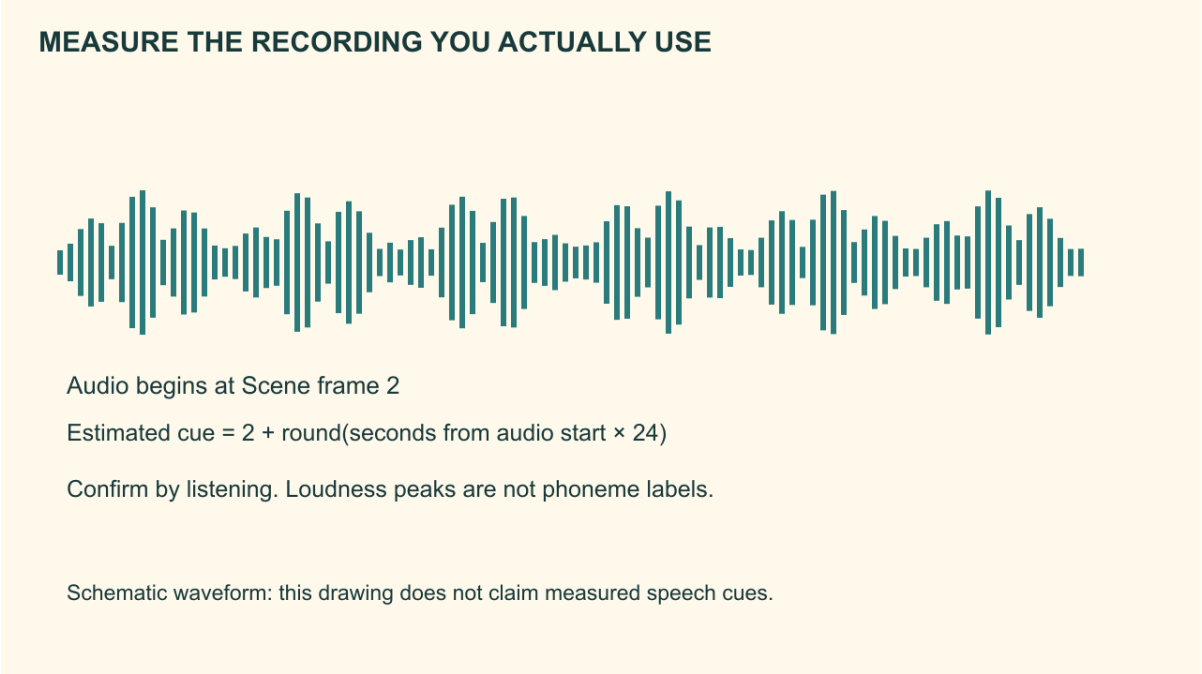
SELECT: A recording application approved in your classroom; then Animate's Library.

TOOLS / ROUTE: Record → trim → export PCM WAV; File → Import → Import to Library.

1. Record "Me count! One… two… three! Me clever!" with a clear pause between numbers, or use the teacher's actual supplied take. Listen to the whole recording before using it.

2. Trim only unwanted silence, keeping a little lead-in. Export a standard PCM WAV and give it a meaningful name. Preserve the original take.

3. Import that recording. The supplied synthetic test sound can test the mechanism, but it is not a natural performance approved for the film.



Cue-sheet method diagram. The waveform is schematic; students measure and listen to their actual take.

WHY · The recorded performance determines cue timing and emotional rhythm.

CHECK / SUMMARY · The file plays clearly with the intended words and no clipped start or end.

FIX · If speech is distorted, rerecord at a lower input level. If the import fails, convert the audio to a supported PCM WAV with the classroom's approved audio tool.

GLOSSARY · clipping: Distortion caused by an audio level exceeding the recording limit.; PCM WAV: An uncompressed audio file format useful for an editing workflow.

DIFFERENT-OBJECT EXAMPLE · A narrator's pause before a reveal affects the edit; the edit should respect the actual take.

STRETCH · Record two contrasting performances and choose based on the story ending.

02 · Place Stream sound and measure the cues

SELECT: Me_clever main Scene; Audio layer frame 2.

TOOLS / ROUTE: Insert Keyframe at 2; Properties → Frame → Sound → Sync: Stream.

1. Create a dedicated Audio layer and place the recording on frame 2. Set Sync to Stream in Frame properties. Measure the audio end, add a short ending hold, and extend the Audio layer AND every visible layer to that same final frame. An early-ending layer must not make a body part disappear.

2. Scrub and listen. On a guide Notes layer record the frame where each phrase/number begins, the strongest vowel and the close or silence.

3. For a timestamp t seconds from the start of the audio, a first estimate is frame 2 + round(t × 24). Confirm by listening; waveform peaks alone do not identify phonemes.

WHY · Stream sound supports timeline synchronization. Real cues must come from this take, not the old 1/25/49 control test.

CHECK / SUMMARY · All words play in the intended order, the clip is not cut off and the cue sheet names actual observed frames.

FIX · If the start is silent, confirm the sound key is at 2 and that enough frames exist. If scrubbing is unreliable, test playback and inspect the selected frame's sound settings.

GLOSSARY · Stream sound: Sound synchronized to timeline playback.; cue sheet: A record linking heard events to timeline positions.

DIFFERENT-OBJECT EXAMPLE · A clap can be estimated from a waveform spike, then confirmed by listening at the marked frame.

STRETCH · Mark the phrase stress separately from the mouth changes.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

03 · Build and align the four-pose mouth bank

SELECT: A new Graphic named Mouth_Poses.

TOOLS / ROUTE: Library → New Symbol → Graphic; internal frames 1–4.

1. Inside Mouth_Poses, put closed at internal frame 1, round at 2, wide at 3 and teeth at 4, with one keyframe per pose.

2. Align all mouths to the same visible lip centre. Source PNG canvases have different transparent margins, so matching their raw X/Y values is not enough.

3. Place one Mouth_Poses instance on a separate Mouth layer over the original caveman_base beard's mouth area in Me_clever. Keep the body on its own locked layer.

WHY · A shared mouth centre prevents lip sync from becoming positional jitter.

CHECK / SUMMARY · Cycling through all four internal frames changes the mouth shape while its visible centre stays in place.

FIX · If the mouth jumps, correct alignment inside the bank. Do not compensate by moving every main-timeline mouth instance separately.

GLOSSARY · mouth bank: A symbol containing a set of reusable mouth drawings.; visible alignment: Matching the artwork itself rather than transparent canvas edges.

DIFFERENT-OBJECT EXAMPLE · Four eye drawings should align by the pupil/eyelid position, not by image canvas corners.

STRETCH · Add a neutral hold using the existing closed pose before creating any extra mouth artwork.

04 · Set manual Single Frame lip-sync keys

SELECT: The placed Mouth_Poses instance on the main Scene timeline.

TOOLS / ROUTE: Properties → Object → Looping: Single Frame; Frame picker or First.

1. First initialize the main Mouth layer at frame 1: select Mouth_Poses, choose Single Frame and First 1 (closed). This prevents looping during the lead-in. At the first speech cue insert a new Mouth keyframe and choose its required pose with Frame picker or First.

2. Add keys at meaningful sound-shape changes. Use round for rounded vowels, wide for broad/open vowels, teeth where the stylized drawing fits and closed for silence or closed lips.

3. Replay in short phrases with sound. Prefer clear selected shapes and holds over a new random mouth on every frame. The four-pose bank approximates speech; it cannot draw every phoneme exactly.

WHY · Single Frame holds the selected internal drawing until the next main-timeline change.

CHECK / SUMMARY · The selected object is Mouth_Poses, not Hand_1. Different main-timeline keys show different First values and corresponding visible mouths.

FIX · If the mouth cycles automatically, choose Single Frame. If all cues show the same pose, check that you inserted main-timeline keys and changed the selected instance at each key.

GLOSSARY · Single Frame: A Graphic playback setting that holds one chosen internal frame.; phoneme: A speech sound; a drawing usually represents a family of related mouth shapes.

DIFFERENT-OBJECT EXAMPLE · A simplified talking fox can hold a wide mouth through a sustained vowel rather than cycling all shapes.

STRETCH · Lead or delay a mouth cue by one frame, compare and justify the clearer result.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

05 · Swap three separate counting hands

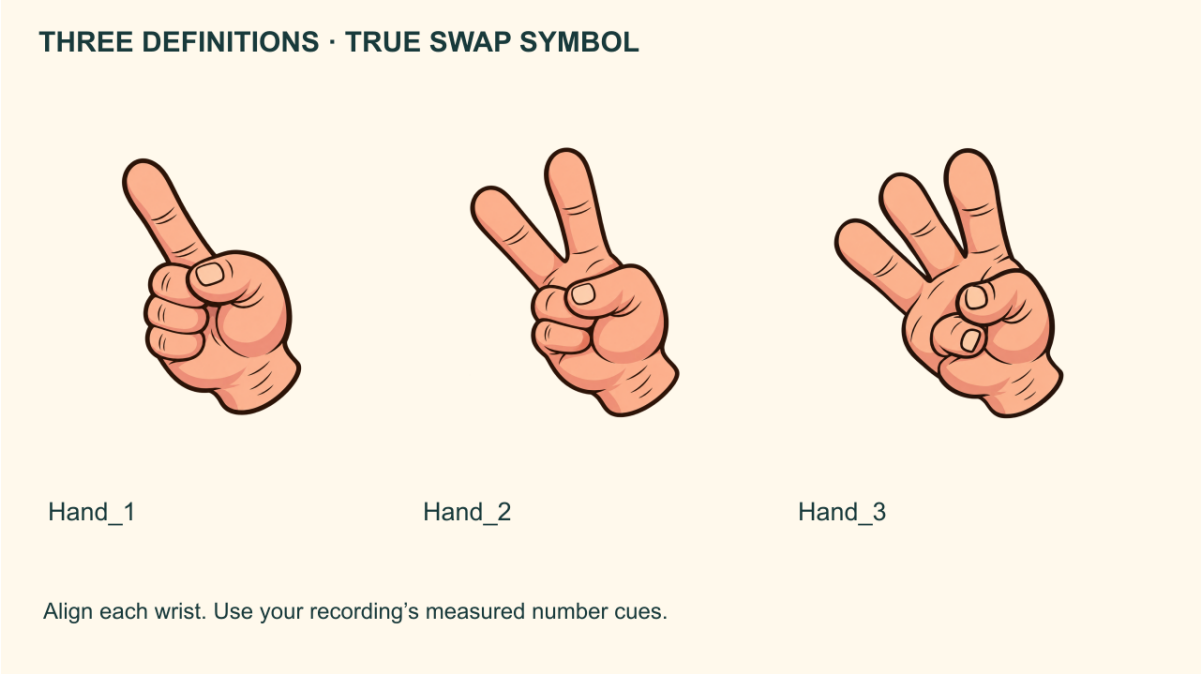
SELECT: The Hand layer in Me_clever; three named hand Graphic definitions.

TOOLS / ROUTE: Modify → Convert to Symbol; Properties → Swap Symbol.

1. Create separate Hand_1, Hand_2 and Hand_3 Graphic definitions from the supplied one-, two- and three-finger images. Align each visible wrist at a common registration location.

2. Place Hand_1 for “one.” Insert a new Hand key at your measured “two” gesture cue, select the instance and Swap Symbol to Hand_2. Repeat for Hand_3 at “three.”

3. Check and correct scale/position after each swap so the wrist remains attached. Keep this as true definition swapping, not a three-frame hand bank disguised as Swap.



Original one-, two- and three-finger artwork belongs to three separate Graphic definitions for true Swap Symbol.

WHY · Gestures communicate counting and can prepare slightly before the emphasized word.

CHECK / SUMMARY · Properties identifies Hand_1, then Hand_2, then Hand_3 at the measured cue keys; the wrist stays fixed.

FIX · If the hand jumps, align the wrist inside each definition and recheck the placed instance transforms. Do not use arbitrary 1/25/49 timings for a different recording.

GLOSSARY · Swap Symbol: Replacing an instance’s referenced Library definition.; gesture cue: The chosen moment at which a meaningful hand action occurs.

DIFFERENT-OBJECT EXAMPLE · A traffic light can swap distinct red/green symbols while retaining its mounting point.

STRETCH · Let the hand prepare before the number, then settle during the word.

06 · Add acting without obscuring the count

SELECT: Body/head action and hand transitions in the counting close-up.

TOOLS / ROUTE: Pose keys; held poses; phrase playback.

1. On “Me count,” establish focus on the stones. During the count keep each finger display readable long enough to understand.

2. For the original flattened seated setup, select the body, mouth and hand layers’ matching frames and Copy Frames into a new Graphic named Count_Performance, preserving the separate inner layers and all cue keys. Verify the copied keys and clear the original visible body, mouth and hand exposures on the main Scene so no duplicate remains beneath the container. Keep Audio on the main Scene. Place Count_Performance from main frame 1 with Play Once/First 1. On “Me clever,” use a restrained whole-container tilt or upward accent and settle, so the body, mouth and hand remain attached. On every added outer key at main frame N, set First to N because this container starts at main frame 1; this preserves the inner cue timing instead of restarting it. Do not claim an independently rotating head on this baked base.

3. Blink only if you have a genuine separate eye/eyelid setup. Do not paste another pair of eyes over baked eyes and call it a finished blink rig.

WHY · Secondary acting should reinforce the words and the result of the story.

CHECK / SUMMARY · The viewer can count all three fingers/stone beats and read pride at the end.

FIX · If everything moves on every syllable, remove unnecessary keys. Preserve one main emphasis per phrase.

GLOSSARY · phrase emphasis: The strongest intended meaning or stress within a spoken phrase.; moving hold: A restrained living pose that preserves the main idea.

DIFFERENT-OBJECT EXAMPLE · A speaker can hold a pointing gesture through an important noun while a small head movement supplies life.

STRETCH · Design an intentionally awkward proud pause after the last word.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

07 · Repair a deliberately mistimed phrase

SELECT: A duplicate lip-sync section, three to five frames out of time.

TOOLS / ROUTE: Select Mouth keys; shift keys; compare with audio.

1. Offset one phrase’s mouth keys enough to make the mismatch visible. Leave the audio fixed.

2. Replay, identify whether the mouth leads or lags, then restore alignment using the actual heard cue.

3. Retain one before/after frame pair with the cue-sheet entry and explain why still images alone cannot prove synchronization.

WHY · The repair tests listening and timing, not just control settings.

CHECK / SUMMARY · Playback of the corrected phrase agrees with the recording and the cue sheet.

FIX · If correcting one word breaks the next, select only the intended keys and preserve their internal rhythm.

GLOSSARY · lead/lag: An event occurring before/after the intended synchronized event.

DIFFERENT-OBJECT EXAMPLE · A door slam picture and sound must be checked in playback even when their timeline keys look close.

STRETCH · Compare a held vowel with excessive mouth chatter and explain which reads better.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

08 · Direct a new vocal reading

SELECT: Your director recording and duplicate counting Scene.

TOOLS / ROUTE: Import take; Stream sound; remeasure cues; mouth and hand keys.

1. Record the changed emotional reading of the same words. Replace the audio in the duplicate and measure its cues again.

2. Revise mouth selections, hand swaps and the final acting accent to match the new take. Do not slide the old key pattern blindly.

3. Save, close and reopen, then listen to the entire Scene. Keep the recording, cue sheet and a playable preview together.

WHY · A new performance requires new synchronization and acting decisions.

CHECK / SUMMARY · The selected performance is intelligible, synchronized and consistent with the film's ending.

FIX · If the new take changes length, extend or trim all relevant Scene layers and the ending hold deliberately.

GLOSSARY · performance timing: The rhythm supplied by a particular delivery of the line.

DIFFERENT-OBJECT EXAMPLE · Independent task: no cue sheet is supplied for a recording that does not yet exist.

STRETCH · Explain why one held mouth or delayed hand gesture supports the emotional reading.

Your checkpoint: predict → observe → revise

Prediction: _____

Observed result / frame: _____

Revision or next check: _____

Keep a small, useful evidence record

At frame ___, I chose ___ so the viewer could ___.
The problem was caused by ___. I changed ___ and checked ___.
Under the changed condition, I revised ___ because ___.

One repair

One independent decision and its trade-off

Delivery check

The selected file reopens. The result plays as intended. Joints/contact and audio were checked where relevant. The changed condition is solved. Required production files and a brief defence are retained.

Method references and attribution

A1 · Adobe Animate: symbols and instances · Adobe reference

A2 · Adobe Animate: timeline layers · Adobe reference

A3 · Adobe Animate: shape tweening · Adobe reference

A4 · Adobe Animate: parenting view · Adobe reference

A5 · Adobe Animate: sound · Adobe reference

A6 · Adobe Animate: camera · Adobe reference

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